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The career and family choices of women: A dynamic analysis of labor force participation, schooling, marriage, and fertility decisions

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Abstract

This paper formulates and estimates a discrete time, discrete choice dynamic labor supply model in which marriage, fertility, and education are choice variables. The dynamics of these choices are captured by various forms of state and duration dependence. Uncertainty comes from the imperfect control women have over births and from a choice-specific random shock to utility each period. Women choose different career and family life-cycle paths because of these uncertainties and also because they have different tastes. The structural parameters of the model are estimated using maximum likelihood estimation techniques with data from the National Longitudinal Survey of Youth.

JEL classification: J12; J13; J16; J22

Keywords: Economics of gender; Time allocation and labor supply; Marriage; Fertility

1. Introduction

Demographic trends and economic theory have led to a growing awareness that women's marriage and fertility choices are closely interrelated with their labor supply decisions and should therefore not be considered exogenous to each other (van der Klaauw, 1996). Many previous studies of female labor supply have avoided modeling this endogeneity by assuming that marriage and childbearing decisions are completed early in life. Some of these studies, for

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example, consider only married women (e.g., Mincer, 1962; Heckman, 1980; Hausman, 1981; Moffitt, 1984; Mroz, 1985) and others condition on completed childbearing (e.g., Cain, 1966; Heckman, 1974; Schultz, 1978; O'Connell, 1990; Klerman and Leibowitz, 1994). Similarly, there is a sizeable literature that examines the influence of children already born on marriage decisions (e.g., Becker et al., 1977; Cherlin, 1977; Waite et al., 1985; Bennett et al., 1989; Waite and Lillard, 1991). However, conditioning on completed choices may result in a sample selection bias. This approach is particularly inappropriate for young women as none of these decisions can reasonably be considered predetermined.

Moreover, empirical research has illustrated the importance of jointly modeling women's labor supply and family choices. Labor supply models that account for the endogeneity of fertility, for example, produce sometimes dramatically different results than studies that do not (e.g., Waite and Stolzenberg, 1976; Schultz, 1978; Rosenzweig and Wolpin, 1980) with no clear pattern of the impact that controlling for endogeneity has on results (Browning, 1992; Orbeta, 2005).[1] A common method of accounting for this endogeneity is to instrument for fertility.[2] However, estimates will be inconsistent if the instrument is not exogenous, and finding truly exogenous variables is difficult (Willis, 1987; Browning, 1992; Nakamura and Nakamura, 1992). Similarly, studies that estimate simultaneous equation models, for example of marriage and fertility (e.g., Koo and Janowitz, 1983; Lillard, 1993; Lillard and Waite, 1993), may result in simultaneous equation bias because of difficult identification issues. Econometric problems such as these are likely to increase substantially as the system of choices becomes more complex. However, these problems can be avoided by viewing choices as jointly determined (Mincer, 1963; Moffitt, 1984).

Jointly modeling labor supply and family choices also makes it possible to disentangle and thus understand the true impact that these decisions have on each other. Disentangling these effects is especially important in the evaluation of policy experiments because policies targeting one choice will likely affect the others as well. Ignoring this interrelatedness or simply conditioning on completed choices may lead to incomplete or inaccurate predictions. Nonetheless, relatively few studies have jointly modeled labor supply and marriage decisions (McElroy, 1985; Johnson and Skinner, 1986, 1988; Haurin, 1989; van der Klaauw, 1996), labor supply and childbearing choices (Cain and Dooley, 1976; Hotz and Miller, 1988; Blackburn et al., 1993), or marriage and childbearing decisions (Koo and Janowitz, 1983; Lillard and Waite, 1993; Brien et al., 1999; Upchurch et al., 1999).

Much of the work that has been done to jointly model women's labor supply and family choices has been within a one-period framework in which agents are assumed to act as if their decisions today have no impact on their future economic environment. This makes it difficult to connect realized decisions to changes in the economic environment over time. Moreover, single-period models cannot properly explain dynamic phenomena such as birth or marriage timing and spacing or the impact of career or family decisions on future earnings. Dynamic models, in comparison, more adequately capture the interdependence of labor supply, education, marriage, and fertility decisions over time.

This paper presents a dynamic female labor supply model with education, marriage, and fertility explicitly modeled as choice variables. Each period women maximize the present value of their utility over a known finite horizon by choosing whether to work in the labor market, be mar-

[1] Research has similarly shown that whether marriage is assumed endogenous or exogenous to labor supply can have a substantial impact on results (e.g., McElroy, 1985; Johnson and Skinner, 1986; Haurin, 1989; van der Klaauw, 1996).

[2] See Browning (1992) for a review of this literature.

ried, use contraception, and attend school. Women account for the impact their current career and family decisions have on their future choices and environment. For example, birth events are stochastically determined conditional on the contraception and marriage choices that women make, and future wages depend upon previous employment and education decisions. Uncertainty in the model comes from the imperfect control women have over fertility and from choice-specific random shocks to utility in each period. Women choose different career and family life-cycle paths because of this uncertainty as well as heterogeneity of preferences.

I use maximum likelihood estimation techniques and 18 years of data on young women from the National Longitudinal Survey of Youth (NLSY) to obtain structural estimates of the model. Estimation is based on a recursive numerical solution of reservation values and expected value functions and is complicated because incorporating childbearing, marriage, and education decisions as choice variables substantially expands the state space. Because of these complications, I employ techniques to reduce the size of the state space and interpolation methods to limit the number of times the value functions need to be calculated.

To date, only a handful of microeconomic papers have jointly modeled labor supply and family choices in a dynamic framework. Hotz and Miller (1988) focus on the interdependence between women's life-cycle fertility and labor force participation decisions; Eckstein and Wolpin (1989) address the human capital investment dimension of women's labor force participation decisions; van der Klaauw (1996) jointly models women's labor force participation and marriage decisions[3]; Keane and Wolpin (1997) focus on the human capital investment dimension of men's education, labor supply, and occupational choice decisions. The only paper that has considered women's labor supply, schooling, marriage and fertility decisions simultaneously in a dynamic framework is Keane and Wolpin (2002). However, Keane and Wolpin do not provide a structural estimation. They instead approximate decision rules using reduced form specifications (which allows them to work with a larger state space).[4]

Macroeconomists have also recently become interested in the dynamics of labor supply and family issues such as fertility and marriage. Greenwood et al. (2003) and Caucutt et al. (2002) model marriage, labor supply, and the quantity and quality of children. Conesa (2000) and Mullin and Wang (2002) look at the timing of fertility decisions and parents' related labor supply and schooling choices. Jones et al. (2003) model the human capital investment decisions of both married and single people (where marital status is exogenous). These macroeconomic papers differ from my paper by modeling choices in a general rather than partial equilibrium framework and by using macro level data to calibrate the model rather than employing maximum likelihood estimation to obtain structural parameter estimates.[5]

I use the structural parameter estimates of my model to predict differences in the life-cycle patterns of marriage, employment, education, and birth control due to differences in observable characteristics, children, and various forms of state and duration dependence. The estimates reveal statistically significant variation in the utility flows associated with women's career and

[3] The work I present in this paper has many similarities to van der Klaauw (1996). However, van der Klaauw does not incorporate schooling and fertility as choice variables. Instead, fertility is endogenous only through its dependence on marital status, and education is assumed to be completed prior to the first period.

[4] I will expand upon how our estimation strategies differ when I present the estimation strategy used in this paper in Section 5.

[5] Fernandez et al. (2004) also model marriage and labor supply decisions in a dynamic macroeconomic framework. The authors do not calibrate their model, instead using regression analysis to explore the intergenerational impact of growing up in a family in which the mother works on the female labor force participation rate.

family choices across observable characteristics such as age, race, ability, and religion. Interesting findings of this portion of the analysis include a negative duration dependence in the preference for marriage. Moreover, results reveal that the nonpecuniary benefits to entering the work force or reentering school after a period of absence outweigh the nonpecuniary costs. Additionally, results suggest that children under the age of 6 have a bonding influence on marriage and that the presence of children increases the utility flows from using contraception.

I then conduct counterfactual simulations to examine the impact on women's career and family choices of changing the return to education, divorce laws, and the cost of childcare. Key results of these policy experiments are that women would stay in school longer, work in the labor market more, spend a smaller fraction of their lives married, and use birth control more frequently if the return to education was increased. In addition, reducing the divorce cost would decrease the number of years in which women are married, but would not substantially affect their labor market work, schooling or birth control decisions. Finally, in response to a childcare subsidy, women would work in the labor market more, spend a greater fraction of their lives married, use birth control less frequently, and spend less time in school.

2. Economic model

In this section, I present the lifetime, constrained utility maximization problem faced by all women. In every period until period T, each woman maximizes the present value of her utility by choosing whether to work, whether to be married, whether to use contraception, and whether to attend school. Let $p_t = 1$ if a woman participates in the labor force in period t and zero otherwise; let $m_t = 1$ if she is married in period t and zero otherwise; let $b_t = 1$ if she uses birth control in period t and zero otherwise; and let $e_t = 1$ if she attends school in period t and zero otherwise.[6] A woman's choice set is denoted by $K_t = \{p_t, m_t, b_t, e_t\}$, where there are at most 16 choices available each period. Note that while women make decisions through period T, they are assumed to live until period T^D, where $T < T^D$.[7]

The utility women derive from their choices each period is comprised of three parts: nonpecuniary utility, pecuniary utility, and a choice-specific random shock to utility.[8] The first component, nonpecuniary utility (q), is the portion of utility that is independent of consumption. Women derive nonpecuniary utility from a variety of sources. First, nonpecuniary utility depends upon the number and ages of children ever born to a woman. Let $n_t = 1$ if a woman gives birth in period t and zero otherwise. Define $N_t^1 = \sum_{\tau=0}^{5} n_{t-\tau}$ as the number of children ever born, as of time t, who are under the age of 6, and $N_t^2 = \sum_{\tau=6}^{18} n_{t-\tau}$ as the number of children ever born who are between the ages of 6 and 18. Finally, let $N_t = N_t^1 + N_t^2$.

Second, various forms of state dependence are captured in nonpecuniary utility. If a woman chooses to work in a given period, her nonpecuniary utility includes her labor force participation

6 It is assumed that only one type of contraceptive method exists. Thus, a woman chooses only whether or not to use birth control. Additionally, the model does not allow for part-time work. Although part-time work is an important option for women, with as many as 25 percent of employed women working part-time, allowing for both part-time and full-time work would double the size of my state space. Therefore, I follow Swann (2005), van der Klaauw (1996), Eckstein and Wolpin (1989), and Hotz and Miller (1988) and assume that women can only work full-time.

7 Allowing for the extra years (T through T^D) increases the time horizon without substantially increasing the number of computations required to estimate the model. Moreover, similar to Wolpin (1984) and Eckstein and Wolpin (1989), all women are assumed to be fecund until time T and infertile from T through T^D.

8 The econometric specification of the nonpecuniary utility function is presented in Section 4.

decision in the previous period to reflect a potential nonpecuniary job finding cost. If she chooses to attend school, her school decision in the previous period is included to capture a possible nonpecuniary school reentry cost. Finally, if she chooses to not be married in a given period, her marriage decision in the previous period is included to reflect a potential nonpecuniary divorce cost.

Third, nonpecuniary utility includes three sources of duration dependence. Let L_{t-1} represent accumulated labor market experience at time $t-1$, M_{t-1} marriage duration at time $t-1$, and H_{t-1} the highest grade completed at time $t-1$. Given this notation, the duration vector, d_{t-1}, is specified as:

$$d_{t-1} = \begin{pmatrix} L_{t-1} \\ M_{t-1} \\ H_{t-1} \end{pmatrix} \tag{2.1}$$

where its elements evolve according to:

$$L_t = L_{t-1} + p_t, \tag{2.2}$$

$$M_t = m_t \cdot [M_{t-1} + m_t], \tag{2.3}$$

$$H_t = H_{t-1} + e_t. \tag{2.4}$$

If married, marriage duration may pick up a woman's nonpecuniary utility of shared memories or marriage-specific capital (excluding children). If working, job tenure may capture the value she derives from good coworker relations, or the increasing disutility of working with previous work effort. If in school, accumulated education may capture the increasing or decreasing effort cost of school.[9]

The second component of the period utility function, pecuniary utility, is the consumption of a composite commodity, c_t. The amount of the composite good a woman consumes each period equals her total available income, less expenditures on children, post-secondary education, and birth control. Consider a woman's three potential sources of income. First, she receives a welfare payment if she is single with at least one child under the age of 18. Second, she earns wage income if she chooses to work in the labor market. Finally, some portion of her husband's wage earnings is available to her if she is married.[10]

The last component of the period utility function, the vector of choice-specific random shocks to utility, $\epsilon(p_t, m_t, b_t, e_t)$, depends upon choices and captures stochastic changes in the utility of each choice. These choice-specific shocks can be interpreted as random variation in the compatibility between choices, such as working and being married, or using contraception and being single. The shocks associated with choices in which a woman is married can be thought of as new draws of marriage match quality. Once married, new draws of these stochastic elements may lead to divorce. Moreover, the shocks associated with choices in which a woman is working can be interpreted as random variation in her wage earnings. Lastly, the shocks associated with choices in which a woman chooses to use birth control may capture random variation in the effectiveness of birth control or a woman's innate fecundity.

[9] Increases in both work experience and education increase earnings profiles. The nonpecuniary values of work experience and accumulated education are specified net of the value of such wage increases.

[10] All (potential) husbands are assumed to work full-time in the labor market.

Given the above notation and assuming β represents a woman's subjective discount factor, X_t her observable characteristics, and E the expectations operator, the objective of each woman is to maximize

$$E\left[\sum_{t=1}^{T}\beta^t U(p_t, p_{t-1}, m_t, m_{t-1}, b_t, e_t, e_{t-1}, c_t, N_t, d_{t-1}, X_t)\right] \tag{2.5}$$

over her choice set where the period utility function is specified as

$$U(\cdot) = q(p_t, p_{t-1}, m_t, m_{t-1}, b_t, e_t, e_{t-1}, N_t, d_{t-1}, X_t) + c_t + \epsilon(p_t, m_t, b_t, e_t), \tag{2.6}$$

and maximization is subject to the budget constraint,

$$\begin{aligned} c_t + \rho^e e_t = m_t \cdot &\left[\psi \cdot \left(w_t^h + p_t w_t - \sum_{\tau=0}^{18}\rho^n n_{t-\tau} - \rho^b b_t\right)\right] \\ &+ (1 - m_t) \cdot \left[1(N_t > 0) \cdot Q_t + p_t w_t - \sum_{\tau=0}^{18}\rho^n n_{t-\tau} - \rho^b b_t\right], \end{aligned} \tag{2.7}$$

where ρ^e denotes the cost of post-secondary education, ρ^n is the per child expenditure cost for children age 18 and younger, ρ^b denotes the cost of birth control, and $1(\cdot)$ is the indicator function.[11] Additionally, Q_t represents the period t welfare payment a woman receives if she is single with at least one child under the age of 18, and w_t represents her wage earnings if she chooses to work in the labor market.[12] A woman's wage earnings are a function of her observed characteristics, characteristics of her local labor market, X_t^L, work experience, and education:

$$w_t = w\big(X_t, X_t^L, L_{t-1}, H_{t-1}\big). \tag{2.8}$$

Lastly, w_t^h denotes her husband's wage earnings if she chooses to be married, and ψ is the fraction of her family's pooled income (less expenditures on family public goods including children and contraception) available to a married woman for her private consumption.[13] In the marriage literature, this fraction is assumed to depend on the sharing rule defined by a Nash solution to a "marriage bargaining" game (McElroy and Horney, 1981; McElroy, 1994). I am not able to identify the parameters of the sharing rule due to data limitations.[14] Therefore, I assume that one-half of pooled family income is available to each spouse for his or her private consumption.[15] Note that because I assume that a husband's wage earnings

[11] Note that the per child cost does not vary with the ages or spacing of children. Thus, economies of scale are ignored. Hotz and Miller (1988) find that while the maternal time inputs in the care of children decline with age, parental expenditures on children do not significantly vary with age. Moreover, note that the per child expenditure cost does not vary with the number of children. Thus, this model cannot say anything about the tradeoff between the quantity and quality of children. Finally, I assume that parents are only financially responsible for children who are 18 years old or younger.

[12] As further discussed in Section 3, the welfare benefit is specified as a fixed guarantee that varies with the number of children a woman has less a fraction of her wage earnings.

[13] The shared cost of family public goods represents a gain to marriage.

[14] Browning et al. (1994) show that identification of the sharing rule requires data on the assignment of at least one private good to a particular spouse. I do not have such data. Moreover, I am not able to model both sides of the marriage decision because there is not sufficient data on the characteristics of spouses and potential spouses.

[15] The choice of one-half is arbitrary. McElroy (1994) shows that while marriage markets yield unique solutions for who marries whom, they do not uniquely identify the distribution of the value of marriage. Instead, there is a set of possible equilibrium allocations called the core.

depend upon the characteristics of his wife (as will be further discussed in Section 4), pooled family income reflects a woman's position in the marriage market even though the sharing rule does not.

Uncertainty in the model comes from the choice-specific random shocks to utility each period and from the imperfect control women have over births. That is, women can only imperfectly control the probability of having a child each period by choosing whether or not to use contraception. The probability that a woman gives birth in period $t+1$, P_{t+1}^n, depends upon her choice of whether or not to contracept in period t, her age, τ, and her marital status in period t:[16]

$$P_{t+1}^n = \Phi(b_t, \tau, m_t), \tag{2.9}$$

where $\Phi(\cdot)$ is the standard normal distribution function.[17] Women make different career and family decisions over their lifetimes because of these two types of uncertainty and also because they have different tastes, as captured by variation in their observable characteristics.[18]

Note that instead of assuming that all women have a marriage offer each period, a more realistic specification might include a marriage opportunity arrival function to allow for uncertainty. As noted by van der Klaauw (1996), the parameters of such an arrival rate are identified only on functional form. Unfortunately, estimation efforts undertaken for this paper to incorporate a marriage arrival rate were unsuccessful, as the parameters of this function proved to be too weakly identified. Consequently, this is left for future work.

Assume that perfect financial capital markets exist and that marriage does not affect an individual's ability to save and borrow.[19] Given these two assumptions and the separability of consumption, the above constrained optimization problem reduces to a lifetime wealth maximization problem modified by the psychic value of the nonpecuniary components of utility including children, marriage, and leisure. This lifetime utility maximization problem can be expressed using value functions. The value function for a particular choice gives the lifetime discounted value of that choice, or "alternative," for a given set of state variables, $S(t)$. State variables include p_{t-1}, m_{t-1}, N_t, d_{t-1}, X_t, X_t^L and $\epsilon_{1t}, \ldots, \epsilon_{Kt}$. In general, the value function at time t given $S(t)$ can be expressed:

$$V_t = V\big(S(t), t\big) = \max_{k \in K} V_k\big(S(t), t\big) \tag{2.10}$$

where

[16] The birth probability function expresses the biological probability that a woman will bear a child. Allowing fertility to depend on marital status captures the increase (or decrease) in sexual activity during marriage, the decrease (or increase) in the degree of carefulness, or the increase (or decrease) in the willingness to have a child.

[17] While the NLSY collects data on abortions, these data are not reliable (Jones and Forrest, 1992). Thus, abortions are not directly modeled. However, the shocks associated with a woman's birth control choices may capture her willingness to terminate a pregnancy.

[18] One might consider incorporating unobserved heterogeneity to the model to explore its effects on women's tastes for labor market work, education, marriage, and contraception use. Adding unobserved heterogeneity would substantially increase the computational cost of estimating the model presented in this paper. Therefore, this endeavor is left for future research. However, note that the estimated duration effects would be smaller if unobserved heterogeneity were present in the model and statistically significant. This is because without unobserved heterogeneity the persistence in women's choices due to unobserved preferences is attributed to duration (Heckman and Singer, 1984). Thus, modeling the unobserved heterogeneity would reduced the estimated duration effects.

[19] For example, an individual who enters a marriage with \$X in savings is able to keep her \$X in savings in the event of a divorce.

$$V_k\big(S(t),t\big) = R_k\big(S(t),t\big) + \beta E \max_{k\in K}\big[V\big(S(t+1),t+1\big),S(t)\big] \quad \forall t \leqslant T-1, \tag{2.11}$$

$$V_k\big(S(T),T\big) = \sum_{\tau=T}^{T^D} \beta^{\tau-T} R_k\big(S(\tau),\tau\big), \tag{2.12}$$

where R_k is the resulting utility level associated with choice k after the budget constraint in Eq. (2.7) is substituted into the utility function in Eq. (2.6), and where the expectations are taken over $\epsilon(p_t, m_t, b_t, e_t)$ and births. Each period, a woman draws values for each choice-specific shock. Given these draws, she chooses the available state with the highest realized value.

3. Data

Data used for this study are from the National Longitudinal Survey of Youth (NLSY), a nationally representative sample of 12,686 young men and women ages 14 to 21 when first interviewed in 1979. Annual, personally administered interviews document their longitudinal experiences as they finish school, make labor market and continuing education decisions, marry, and start families of their own.[20] This study uses data from 1979 through 1996 and samples from the 3108 women who comprise the core, nationally representative subsample.[21] From this group, 941 women who were between the ages of 14 and 17 in 1979 were selected.[22]

The NLSY is well suited for this study. First, the survey begins tracking some individuals at a sufficiently young age to reasonably assume that marriage, employment, continuing education, and family planning decisions are not made before the sample panel period, thus avoiding the unobserved initial conditions problem. Second, detailed information is collected on the timing of employment spells, birth events, schooling, and marriages. Third, the NLSY contains information on the contraception choices of women.

In any given year, a woman is defined to be working if she is employed for 80 or more hours per month in at least six months. She is defined to be married if she is married before the first of July. Furthermore, she is defined to be in school if she is enrolled during both the fall of the previous year and the spring of the current year.[23] In 1982, the NLSY began asking female respondents whether they use birth control and, if so, the frequency with which they use it.[24] A woman is defined to be using birth control in a given year only if she reports that she always uses birth control.

Based on the assumption that all market work is full-time, a woman's wage earnings are defined as her hourly wage rate times 2000 hours.[25] I use the reported hourly wage for the current or most recent job to estimate a female wage equation, which I then use to predict wages

[20] The NLSY switched from annual interviews to biennial interviews after 1994.

[21] The NLSY data include a supplemental subsample that oversamples Hispanics, blacks, and economically disadvantaged whites, and a military subsample.

[22] 58 percent of women in the nationally representative subsample were rejected because they were 18 years of age or older in 1979, 33 respondents were rejected because they gave birth before 1979, and 69 women were rejected because they reported never having entered the 10th grade. All other respondents were rejected due to missing data.

[23] Moreover, she must be enrolled "full-time," where "full-time" means she is enrolled in high-school and is under the age of 20 or she is enrolled in college and is classified by her college as a full-time student.

[24] Female respondents were questioned on their use of birth control in 1982, 1983, 1984, 1985, 1986, 1988, 1990, 1992, 1994, and 1996. The appendix discusses how I simulate birth control choices for the years in which birth control data were not available.

[25] This is the same approach taken in van der Klaauw (1996) and Swann (2005).

for all women.[26] I similarly construct a (potential) husband's wage earnings by multiplying his hourly wage rate by 2000 hours. The NLSY does not collect sufficient information on the characteristics of spouses to estimate a wage equation for (potential) husbands using the sample of women. Therefore, I draw a random sample of 2073 men from the NLSY to estimate a male wage equation.[27] Table 2 reports the estimates of both wage equations, and estimates are discussed in Section 4.

The NLSY data is supplemented with data on the AFDC benefit to which a single parent family is eligible. AFDC benefits consist of a fixed guarantee that varies with the number of children the parent supports, minus a fraction of the parent's income. The guarantee is set at the state level, while the tax rate on income is determined by the federal government. Although the determination of AFDC benefits appears straightforward, in practice benefits depend on various factors resulting in "effective" tax rates and guarantees that differ from those published. Several authors have estimated effective guarantees and tax rates including Hutchens (1978), Moffitt (1979), Fraker et al. (1985), and McKinnish et al. (1999). This study uses the Fraker et al. estimates for the period 1979 to 1982 and the McKinnish et al. estimates for 1983 to 1991. Furthermore, I forecast effective guarantees and tax rates for 1992 through 1996 by assuming benefit parameters follow an ARMA(1,1) process. Because this study does not model migration, average state effective guarantee levels and tax rates are used for all women.[28]

For all other variables used in this analysis, the following definitions apply. The binary variable n_t is defined to equal one in a given year if and only if a woman gives birth before the first of July. "Race" equals one if a woman is white and zero otherwise. "Protestant" is a dummy variable for whether a woman is Protestant, and "Catholic" is a dummy variable for whether she is Catholic. "AFQT" is defined to equal one if the score earned on the Armed Forces Qualification Test is between 0 and 25, two if the score is between 26 and 50, three if the score is between 51 and 75, and four if the score is between 76 and 100.[29]

Table 1 reports descriptive statistics for the sample. The first observation year for each sample member is defined as the year in which she enters the 10th grade, and the last sample year is either 1996 or the year in which she drops out of the sample. Given these definitions, the average sample member is observed for 16 years, resulting in 15,626 "persons-years." Table 1 also indicates that the average woman is a white 24 year old, has graduated high school, and earns \$10.25 an hour.[30] Of the 15,626 person-years, 61 percent are spent working, 20 percent in school, and 47 percent married. Among the 7677 person-years in which contraception information is collected, birth control is used 74 percent of the time.

[26] In cases where the hourly wage rate is missing, annual salary divided by the number of hours worked per year is used instead.

[27] Like the random sample of women, the random sample of men was selected from the core, nationally representative subsample. Of the sample of 3003 men, 930 were rejected due to missing data. While only women between the ages of 14 and 17 in 1979 were selected, men between the ages of 14 and 21 in 1979 were selected to represent the available pool of spouses.

[28] Over the period 1979 to 1996, the average effective guarantee for a family of two is \$281.73, and the average effective tax rate is 36.22 percent.

[29] The Armed Services Vocational Aptitude Battery (ASVAB) is a set of 10 tests given by the military for enlistment screening purposes. In 1980, the ASVAB was administered to over 90 percent of NLSY respondents. The AFQT is a subset of the ASVAB, consisting of the following four tests: paragraph comprehension, arithmetic reasoning, word knowledge, and numerical operations. Scores on the AFQT range from 0 through 100.

[30] Wages are reported in 1983 dollars.

Table 1
Descriptive statistics

Variable	No. of obs.	Mean	St. dev.
Sample of 941 individuals			
Age in 1st period	941	16.508	0.588
White	941	0.879	0.326
AFQT score	941	2.328	1.059
Protestant	941	0.504	0.500
Catholic	941	0.339	0.474
Education in 1st period	941	10.147	0.394
Years in sample	941	16.606	2.136
Sample of 15,626 person-year observations			
Age	15,626	24.444	4.974
Education	15,626	12.748	1.946
Kids ages 1–5	15,626	0.371	0.554
Kids ages 6–12	15,626	0.086	0.281
Kids ages 13–18	15,626	0.006	0.077
Married	15,626	0.470	0.499
Working	15,626	0.613	0.487
In school	15,626	0.199	0.399
Using birth control	7677	0.743	0.437
Wage	8962	10.25	8.038
Spouse's wage	5340	16.51	17.825

Note: Wages are in 1983 dollars.

Table 2
Wage equations

Variable	Women's wage equation	Men's wage equation
Constant	-1.072^{**}	-0.448^{**}
	(0.073)	(0.131)
Age	0.066^{**}	0.071^{**}
	(0.002)	(0.004)
White	0.048	0.257^{**}
	(0.018)	(0.036)
AFQT score	0.049^{**}	0.042^{**}
	(0.006)	(0.010)
Education	0.084^{**}	0.042^{**}
	(0.003)	(0.006)
Experience	0.092^{**}	0.054^{**}
	(0.005)	(0.008)
Experience squared	-0.003^{**}	-0.003^{**}
	(0.005)	(0.001)
Unemployment rate	-0.018^{**}	-0.020
	(0.005)	(0.009)
R^2	0.578	0.266
No. obs.	8962	5340

** Significance at 0.05.

4. Econometric specification of the structural model

This section presents the econometric specification of the economic model in Section 2. Recall that the utility function presented in Eq. (2.6) is the sum of three terms: nonpecuniary utility, consumption, and a choice-specific random error term. For each choice, nonpecuniary utility depends upon a choice-specific constant, observable characteristics including race, religion, AFQT score and age, the nonpecuniary utility flow from children, and the various forms of state and duration dependence discussed in Section 2. Specifically, the nonpecuniary utility function is[31]:

$$q_t = b_{0,k} + b_{1,k} X_t + \zeta_t + b_3 p_t \cdot (1 - p_{t-1}) + b_4 p_t L_t \\ + b_5 e_t \cdot (1 - e_{t-1}) + b_6 e_t H_t + b_7 (1 - m_t) \cdot m_{t-1} + b_8 m_t M_t, \tag{4.1}$$

where ζ_t is the nonpecuniary utility flow from children. This flow depends upon the age distribution of the current stock of children and is specified as:

$$\zeta_t = a_{1,k} N_t^1 + a_{2,k} N_t^2, \tag{4.2}$$

where k indexes choices (marital status, labor force participation, schooling and birth control).[32]

Note that the utility women derive from their children depends upon their choices. For example, the utility a single woman derives from her children differs from the utility enjoyed by a married woman.[33]

The second component of the utility function, consumption, depends upon a woman's choices and her budget constraint. If she works in the labor market, her budget constraint is a function of her wage earnings. Her wage equation (in logs) is:

$$\ln(w_t) = \beta_0 + \beta_1 X_t' + \beta_2 H_{t-1} + \beta_3 L_{t-1} + \beta_4 (L_{t-1})^2 + \beta_5 X_t^L, \tag{4.3}$$

where X_t' is a subset of observed characteristics including AFQT score, age, and race.[34]

If married, a woman's budget constraint also depends upon her husband's earnings. Assortative mating has been documented across a range of characteristics including race, age, education, and religion.[35] Therefore, each (potential) husband's vector of observed characteristics, X_t^h, and his highest grade completed, H_{t-1}^h, are assumed to be identical to those of his wife. This is the same approach taken in van der Klaauw (1996). Note that this implies that women limit their marital search to a pool of potential spouses who are of the same race, age, ability, and education as them. The choice specific shocks to utility for choices in which marriage is chosen pick

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31 A base choice, $\{p_t = 0, m_t = 0, b_t = 0, e_t = 0\}$, is picked to serve as a benchmark for the other 15 choices.

32 The benefit a mother derives from each child does not vary with the number or ages of children within both age groups. Moreover, due to data limitations, only the net benefit of children, $a_{l,k} - \rho^n$ where $l = 1, 2$, can be identified. Similarly, the cost of birth control, ρ^b, cannot be separately identified from the birth control specific constant.

33 Although the "value" of having children depends upon marital status, the model does not distinguish between children born within marriage and children born out-of-wedlock. Consequently, I cannot say anything about the impact of children born out-of-wedlock on the transition into and out of marriage. Further note that the stochastic nature of birth events may, in part, explain out-of-wedlock births.

34 Note that wages depend on total work experience without regard to when that experience was accumulated. Given their primary role in childbearing and rearing, women commonly experience work interruptions leading to intermittent labor force participation. Human capital theory suggests that work disruptions may reduce wages because of skill depreciation or by serving as a signal of low productivity to employers. However, empirical research suggests that while work disruptions do lower wages, wages quickly rebound (Mincer and Ofek, 1982; Light and Ureta, 1995; Baum, 2002).

35 See Johnson (1980), Epstein and Guttman (1984), Mare and Winship (1991), and Pencavel (1998) for examples.

up the extent to which this sorting is not accurate. Additionally, the marriage-specific constant represents the attractiveness of marriage in general as well as the acceptability of specific potential mates. Therefore, it reflects the portion of a (potential) husband's suitability that is not captured by allowing his characteristics to depend on his wife's characteristics. Furthermore, let L^h_{t-1} represent his labor market experience, defined as the difference between his age and his highest grade completed. Given this notation, a (potential) husband's wage equation (in logs) is:

$$\ln(w^h_t) = \alpha_0 + \alpha_1 X^h_t + \alpha_2 H^h_{t-1} + \alpha_3 L^h_{t-1} + \alpha_4 (L^h_{t-1})^2 + \alpha_5 X^L_t. \tag{4.4}$$

Finally, consider the specification of the third component of the utility function, the random shocks to utility associated with each choice k, ϵ_k. I assume that these shocks have independent extreme value distributions:

$$F(\epsilon_k) = \exp\{-\exp\{-\epsilon_k/\tau\}\}, \tag{4.5}$$

with variance $\tau^2\pi^2/6$.[36] Models of this type typically specify errors to be distributed i.i.d. extreme value, since this distributional assumption has two important benefits (Rust, 1987; Berkovec and Stern, 1991; van der Klaauw, 1996). First, it implies that the expected value of the maximum of next period's value functions in Eq. (2.11) has a nice closed form solution. It is straightforward to show that:

$$E \max_{k\in K} V_k = \tau\left[\gamma + \ln\left(\sum_{k\in K} \exp\{\overline{V_k}/\tau\}\right)\right], \tag{4.6}$$

where γ is Euler's constant, and $\overline{V_k} = V_k - \epsilon_k$.

Second, it implies that the choice probabilities have a convenient form. Specifically, each choice probability is multinomial logit. Let $d_{tk} = 1$ if choice k is chosen at time t. The probability of choosing choice k at time t is then:

$$\Pr(d_{tk} = 1) = \frac{\exp\{\overline{V_k}/\tau\}}{\sum_{j\in K} \exp\{\overline{V_j}/\tau\}}. \tag{4.7}$$

It remains only to specify each agent's information set. At each $t = 1, 2, \ldots, T^D$, an agent knows her full set of exogenous characteristics, characteristics of her local labor market, her past choices, and her tenure vector up to time t. In addition, she knows current and past realizations of the shocks to utility associated with each choice. While the future values of these shocks are unknown, she does know the distribution of these errors. Finally, each agent knows her (potential) husband's set of exogenous characteristics.

5. Estimation

The dynamic model is solved by backward recursion. Recall that agents cannot make decisions in periods T through T^D. Therefore, I assume an individual makes the same choice in periods T through T^D as she made in period $T-1$ where k^*_{T-1} denotes this choice. Person i's period T value function is then:

$$V^i_T = E \sum_{\tau=T}^{T^D} \beta^{\tau-T} R^i_{k^*_{T-1}}(S(\tau), \tau). \tag{5.1}$$

[36] That is, the variance is expressed as a multiple of $\pi^2/6$, where τ^2 is this multiple.

With V_T in hand, her value functions for all $t < T$ are evaluated iteratively.[37]

Due to the large size of the state space, solving the dynamic programming problem is not a trivial task. To calculate the alternative-specific value functions at time $t = 0$, the alternative-specific value functions at all future times and at all feasible state points must be computed. It is too computationally expensive to evaluate the value functions for all sample observations at all possible state points. Therefore, I employ two techniques to reduce both the size of the state space and the number of required computations.

First, I place upper bounds on several state variables. Specifically, education can take on only 6 values (grades 10 through 15),[38] labor market experience only 3 values (zero through 2 years), and marriage duration only 3 values (zero through 2 years). Note that this does not imply that more than 2 years of marriage duration, for example, do not matter. Rather, it implies that the impact on utility of 3 years of marriage is the same as the impact of 2 years.[39] In addition, I only track the age of the youngest child. This is because to estimate the model given the utility flow from children as specified in Eq. (4.2), I would have to track the ages of all children ever born to each woman. Since each child increases the size of the state space by a factor of 18, this is not computationally feasible. Given these bounds, there are still $2^8 \cdot 19 \cdot 3^2 \cdot 6 \cdot 28 = 7{,}354{,}368$ possible state space combinations.

Second, rather than estimate the value functions at all possible state space points, I evaluate them at only a subset of points and interpolate them at all remaining points. Let Z denote a vector consisting of the following subset of state variables: education, marriage duration, labor market experience, and age of the youngest child.[40] While there are $6 \cdot 3 \cdot 3 \cdot 19 = 1026$ values of Z, the value functions are computed for only 64 values: 4 out of 6 values of education, 2 out of 3 values of marriage duration, 2 out of 3 values of labor market experience, and 4 out of 19 values of age of the youngest child. Let $\tilde{Z}$ represent a vector of these 64 values of Z. The value functions for all other values of Z are interpolated by taking a weighted average of the value functions at the "closest" state space points, where the closest points are those for which the values of the variables in $\tilde{Z}$ are closest to the interpolated point's values. Weights are inversely proportional to the distance between an interpolated point and its closest points (Stinebrickner, 1996).[41]

Following a weighting scheme similar to that of Brien et al. (2006), let ω_{ij} denote the weight attached to the jth closest point for the ith interpolated point:

$$\omega_{ij} = \left|Z_1^i - \tilde{Z}_1^j\right|^p \left|Z_2^i - \tilde{Z}_2^j\right|^p \left|Z_3^i - \tilde{Z}_3^j\right|^p \left|Z_4^i - \tilde{Z}_4^j\right|^p \tag{5.2}$$

[37] This is the standard approach taken in this literature. For examples, see Eckstein and Wolpin (1989), van der Klaauw (1996), and Keane and Wolpin (1997).

[38] Education was bound at 15 instead of 16 to reduce the computational cost of estimating the model. Thus, I cannot capture any potential graduation, or "sheepskin," effects. While the existence of sheepskin effects is well-documented, estimates based only on an individual's total years of education will be biased because total years of education imperfectly measure degree attainment. This would be particularly problematic for my measure, highest grade completed, which is constructed by summing the years a woman chooses to attend school over time and, thus, may not accurately measure the number of grades actually completed.

[39] Increasing the upper bound on a given state variable would result in a smaller parameter estimate (in absolute value).

[40] Other state variables each period include the four choice variables, the lagged marriage, labor market work and schooling decisions, and whether or not a woman gives birth.

[41] Each interpolated point has 16 "closest" points. For each variable Z_q in Z, there are two closest values in $\tilde{Z}$: one less than or equal to the interpolated point's value and another greater than or equal to the interpolated point's value. Therefore, each interpolated point has $2^4 = 16$ closest points.

where

$$\tilde{Z}_q^j = \begin{cases} \tilde{Z}_{qh}^j & \text{if } Z_q^j = \tilde{Z}_{ql}^j \\ \tilde{Z}_{ql}^j & \text{if } Z_q^j = \tilde{Z}_{qh}^j \end{cases}, \quad \forall q = 1, 4 \tag{5.3}$$

and where $\tilde{Z}_{ql}$ is the lowest closest value of Z_q for the ith interpolated point and $\tilde{Z}_{qh}$ is the highest closest value of Z_q for ith interpolated point.[42] The value function of the ith interpolated point is interpolated by:

$$V_i = \sum_{j=1}^{16} \hat{\omega}_{ij} V_j, \tag{5.4}$$

where

$$\hat{\omega}_{ij} = \frac{\omega_{ij}}{\sum_{j=1}^{16} \omega_{ij}}. \tag{5.5}$$

Given the solution to the dynamic programming problem and data on an individual's choices and state variables, I estimate the parameters of the agent's utility function and budget constraint and her beliefs about the law of motion for the state variables in two stages. In the first stage, I estimate the parameters of the birth probability function presented in Eq. (2.9) and the wage equations specified in Eqs. (4.3) and (4.4). The parameters of these functions are estimated outside the structural dynamic model to reduce the computational cost of the dynamic programming problem and are then treated as given in the estimation of the structural model. This implies that a woman knows with certainty how her wages and the wages of her (potential) husband will evolve over time as well as the impact that her age, contraception choices, and marital status decisions will have on the probability of giving birth.

I estimate the parameters of the birth probability by estimating a probit in which the dependent variable is 1 if a birth occurs and 0 otherwise. These parameters are then used to predict the probability that a woman gives birth within the structural model, conditional on her choice of whether or not to use contraception, her age, and her marital status. The estimation of the parameters of the wage equations was discussed in Section 3. These parameters are used to predict own and (potential) husband's wages within the structural model conditional on a woman's own characteristics, education, labor market experience and characteristics of her labor market.

Given the parameters of the birth probability and wage functions, I estimate the remaining parameters of the structural model in the second stage. Let θ be a vector of the parameters associated with Eqs. (2.6) and (2.7), where θ is estimated by maximizing the likelihood function for the sample of data. This likelihood function is the product over choices, time, and individuals of the probabilities of choosing one alternative over another conditional on $S(t)$. Choice probabilities come from the decision rules. The sample likelihood function is specified as:

$$L(\theta) = \prod_{i=1}^{N} \prod_{t=0}^{T-1} \prod_{k=1}^{K} \Pr\bigl(d_{tk}^i = 1 \mid S(t), d_{t-1,k}, \theta\bigr)^{d_{tk}^i}. \tag{5.6}$$

The likelihood function in Eq. (5.6) is maximized using a Newton-based algorithm. The estimation begins with an initial guess of θ. Conditional on this guess, the likelihood function and its

42 For $p \geqslant 2$, the weights have the nice property that the weighted average is continuous at the seams. For this study, $p = 2$.

derivatives with respect to θ are computed. The derivatives of the likelihood function are used to update θ. This procedure is continued until the likelihood function is maximized. Maximization of the likelihood function yields consistent and asymptotically normal estimates.

I use the Berndt et al. (1974) technique to estimate the standard errors of the parameters.[43] The estimate of the asymptotic covariance matrix is the inverse of the sum over people of the outer product of the partial derivatives of each person's contribution to the likelihood function:

$$\mathrm{Cov}(\hat{\theta}) = \left[\sum_{i=1}^{N}\left[\frac{\partial \log L_i}{\partial \hat{\theta}}\frac{\partial \log L_i}{\partial \hat{\theta}'}\right]\right]^{-1} \tag{5.7}$$

where is $\hat{\theta}$ the simulated maximum likelihood estimate of θ.

Data on the variation in wage earnings across women identify the parameters of the wage equation presented in Eq. (4.3). Data on the variation in earnings across men identify the parameters of the husband's wage equation presented in Eq. (4.4). Variation in the choices across individuals identifies the parameters in Eqs. (4.1) and (4.2). Lastly, variation in the data on birth events across individuals identifies the parameters of the birth probit equation presented in Eq. (2.9).[44]

6. Results

6.1. Parameter estimates

As discussed in Section 5, estimating the structural model is a two-step process. Parameters of the wage equations and birth probability function are estimated first to reduce the computational cost of estimating the full structural model. Table 3 presents parameter estimates of the female and male wage equations specified in Eqs. (4.3) and (4.4), where local labor market characteristics are captured by the unemployment rate. As expected, results indicate that wages increase with age, ability, and education. The first four rows of Table 3 present parameter estimates of the birth probability function. Results reveal that the probability of giving birth declines with age, the use of birth control in the previous period, and is higher if women are married rather than single.

In the second stage, I estimate the dynamic programming model taking the parameter estimates of the wage equations and birth probability function as given. Table 3 presents the structural parameter estimates of this model with the discount factor, β, set to 0.8.[45] The model was estimated with the optimization horizon, T, equal to the age at which a woman enters the

[43] Standard errors are not corrected for the fact that the estimated values for wages and birth probability parameters are used in the second estimation step.

[44] An alternative estimation strategy is employed by Keane and Wolpin (2002), who use Monte Carlo integration to approximate the expected value of the maximum of next period's value functions (the "Emax functions") at a subset of state points and use these values to fit polynomial approximations to the Emax functions. They then use these approximations to simulate the model.

[45] The choice of the discount factor, though consistent with the previous literature, is somewhat arbitrary. Authors such as Eckstein and Wolpin (1989), Berkovec and Stern (1991), and van der Klaauw (1996) have reported difficulty in estimating the discount factor. Keane and Wolpin (1997) estimate a discount factor of 0.78 for their model of the career decisions of young men. Swann (2005) estimates a discount factor between 0.81 and 0.84 for his model of the employment, marriage, and welfare decisions of women. A discount factor in the range of 0.75 through 0.95 is usually chosen.

Table 3
Structural model estimates

Variable	Estimate	Std. error
Birth probability fn.		
Constant	−0.270*	0.172
Age	−0.050**	0.007
Lagged birth control	−0.677**	0.052
Lagged marital status	0.665**	0.055
Working		
Constant, $b_{0,k}$	−1.786**	0.183
White	0.276**	0.079
AFQT	0.051**	0.024
Age	0.004	0.009
Protestant	0.199**	0.067
Catholic	0.143**	0.073
Kids age <6 years old, $a_{1,k} - \rho^n$	−1.206**	0.060
Kids 6 years old and older, $a_{2,k} - \rho^n$	−0.357**	0.079
Labor market experience, b_4	0.775**	0.086
In school		
Constant, $b_{0,k}$	4.077**	0.463
White	−0.651**	0.116
AFQT	0.682**	0.034
Age	−0.417**	0.010
Protestant	0.039	0.085
Catholic	−0.027	0.088
Kids age <6 years old, $a_{1,k} - \rho^n$	−2.774**	0.147
Kids 6 years old and older, $a_{2,k} - \rho^n$	0.271**	0.131
Highest grade completed, b_6	0.706**	0.048
Married		
Constant, $b_{0,k}$	−1.612**	0.204
White	0.185**	0.033
AFQT	−0.022*	0.011
Age	0.049**	0.004
Protestant	0.018	0.031
Catholic	−0.002	0.034
Kids age <6 years old, $a_{1,k} - \rho^n$	1.064**	0.082
Kids 6 years old and older, $a_{2,k} - \rho^n$	0.072	0.048
Marriage duration, b_8	−0.242**	0.122
Using birth control		
Constant, $b_{0,k} - \rho^b$	−1.228**	0.141
White	0.138**	0.058
AFQT	0.166**	0.018
Age	0.067**	0.005
Protestant	0.038	0.053
Catholic	−0.065	0.056
Kids age <6 years old, $a_{1,k} - \rho^n$	0.715**	0.081
Kids 6 years old and older, $a_{2,k} - \rho^n$	0.386**	0.058
School reentry cost, b_5	−2.206**	0.083
Job finding cost, b_3	−2.758**	0.040
Divorce cost, b_7	5.822**	0.374
Cost of college, ρ^e	8.192**	0.262
Marginal utility of consumption	0.060**	0.007

* Significance at 0.10.
** Idem, 0.05.

tenth grade plus 28 years and T^D equal to the age at which she enters the tenth grade plus 50 years. Finally, the variance multiplier τ (from Eq. (4.5)) was set to 1.[46]

The structural parameter estimates of the dynamic model reveal statistically significant variation in the utility flows associated with marriage, contraception, education, and labor supply choices across observable characteristics.[47] These estimates result from an estimation process designed to pick parameter values to match the data as closely as possible. In what follows I offer plausible explanations for what aspects of the data the model attempts to fit by picking particular parameters. Note that the policy implications for each are very different, suggesting the need for further research.

For instance, results indicate that nonwhites enjoy lower utility flows from marriage than whites. This reflects that marriage rates are lower among nonwhites even after controlling for observable characteristics. All else equal, nonwhite women may derive less utility from marriage than white women because they have relatively fewer "suitable" potential spouses due to lower educational attainment and labor force participation rates among minority men (Wilson and Neckerman, 1986; Lichter et al., 1991; Brien, 1997).

Table 3 also shows that nonwhite women derive lower utility flows from using contraception than white women. This reflects that nonwhites use birth control less regularly than whites (Bachrach, 1984; Hogan et al., 1985; Stephen et al., 1988). It has been hypothesized that this difference is driven by access to family planning services (Stephen et al., 1988). Torres and Singh (1986) show that knowledge of various forms of contraception is higher among white teenagers compared to both black and Hispanic teenagers. An alternative explanation is that nonwhite and white women have different cultures regarding contraception use and childbearing. Among Hispanics, for instance, motherhood is highly valued and women attain adult status by giving birth (Cooksey, 1990). It may also be more socially acceptable in the nonwhite community to have children out-of-wedlock; out-of-wedlock birth rates are substantially higher among nonwhites than whites.[48]

Findings further reveal that, all else equal, nonwhites enjoy lower utility flows from labor market work, possibly reflecting wage differences not captured by observed characteristics, nonwage discrimination or unobserved preferences for leisure. Lastly, results indicate that nonwhites derive a higher utility flow from being in school than whites, perhaps because nonwhites receive a higher return to education than their white counterparts. Belman and Heywood (1991) and Ashraf (1994), for example, find that the returns to college completion are higher for blacks than whites.

Next consider the estimated impact of age on the utility flows associated with women's career and family choices. Reflecting the positive relationship observed in the data between age and marriage, the estimated utility gains from marriage are increasing in age. This may be driven by marriage market factors if the probability and/or quality of marriage offers increase with age.

[46] Setting τ to 1 effectively scales utility by τ. Utility can be divided by τ without changing behavior since utility is ordinal.

[47] These estimates indicate the incremental utility flow associated with a particular choice relative to the base choice when the associated covariate is increased by one unit. For example, married whites who use birth control, work in the labor market, and who are not in school receive a utility flow that is $0.185 + 0.138 + 0.276 = 0.599$ units higher than nonwhites (relative to the utility flow of the base choice). Recall that the base choice, $\{p_t = 0, m_t = 0, b_t = 0, e_t = 0\}$, is the alternative in which a woman chooses to not work, not be married, not use birth control, and not be in school. All women have the same utility in the base choice.

[48] In 2002, for example, the out-of-wedlock birth rate was 23 percent for whites, 44 percent for Hispanics, and 68 percent for blacks (CDCP, 2002).

This result may also indicate a positive impact of maturation on marriage or changing tastes for marriage over time. Table 3 also reveals that the gains from schooling are decreasing in age, most likely because the desire for additional years of schooling declines as the number of years women have to reap the benefits decreases. Lastly, I find that the utility flows from birth control are increasing in age. This reflects that contraception use increases with age particularly between the teenage years and early to mid-twenties.[49] Barriers to the use of birth control are thought to be related to various factors such as knowledge of or access to family planning services, the behavior and attitudes of one's peers and sexual partners (Burt, 1995), the stability and predictability of relationships, educational and career opportunities, income, and impulsiveness (Moore, 1995). Such barriers may be particularly high among adolescents (Burt, 1995) and decline with age.

According to standard neoclassical labor supply theory, children (particularly young children) decrease women's labor force participation by increasing the value of time spent at home. Indeed, numerous studies have shown children to be negatively related to various measures of female labor supply.[50] The structural parameter estimates in Table 3 reflect this; children decrease the utility gains from labor market work. Note that this decrease is greater for children under the age of 6 for whom caring is more time-intensive compared to children ages 6 and older. Similarly, children under the age of 6 reduce the utility flows associated with being in school. On the other hand, children ages 6 and older (who are in school themselves) increase these flows.

The parameter estimates further reveal that children increase the utility flows from contraception use. This may reflect that the desire for additional children is declining in the number of children a woman already has (Ryder, 1981; Stephen et al., 1988). Alternatively, it may capture that having children increases a woman's knowledge about her fecundity (Stephen et al., 1988).

Table 3 also indicates that children increase the utility gains from marriage. This accords with economic theory that views children born within marriage as marriage specific capital (Becker, 1973; Becker et al., 1977). Moreover, the estimated impact on utility of children under the age of 6 is statistically significant and larger in magnitude compared to the estimated impact of children ages 6 and older. Indeed, empirical work has found that married couples with young children in particular are less likely to divorce than couples without children (Cherlin, 1977; Waite et al., 1985; Waite and Lillard, 1991). Unfortunately, the data do not allow me to assign the paternity of children, and children from a relationship other than the current marriage may be a source of conflict within the marriage (Cherlin, 1978; Heaton, 1991; Waite and Lillard, 1991). As a result of the inability to model this, the coefficient estimate on the utility flow from children while married may underestimate the true impact that marital children have on an existing marriage.

The estimates imply positive duration dependence in the preferences for labor market work and schooling and a negative duration dependence in the preference for marriage. Lillard and Waite (1993) and Lillard et al. (1995) find that when controlling for both age and marriage duration, the utility flows from marriage increase in age and decrease in the first few years of marriage duration (but level off thereafter). According to Lillard et al. (1995), age and marriage duration are two distinct types of duration. While marriage duration reflects the accumulation of marriage-specific capital as well as the quality of the match, age reflects the impact of maturation. This may similarly explain my findings that the utility flows from marriage are increasing in age and decreasing in marriage duration. However, as discussed in Section 5, I put an upper bound

[49] In 2002, for example, 31.5 percent of women 15–19 years of age used contraception compared to 60.7 percent of women ages 20–24, 68 percent ages 25–29, 69.2 percent ages 30–34, and 70.8 percent ages 35–39 (CDCP, 2004).

[50] Early studies include Mincer (1962), Cain (1966) and Bowen and Finegan (1969). See Lehrer and Nerlove (1986) and Nakamura and Nakamura (1992) for surveys of this topic.

of two years on marriage duration for computational reasons. Therefore, I am not able to say whether the utility flows are decreasing in all years of marriage duration or only in the first few years.[51]

While the estimates reveal a positive duration dependence associated with labor market work and schooling, they reveal a negative state dependence for these choices. As Table 3 indicates, the school reentry and job finding costs are statistically significant and negative, implying that the nonpecuniary benefits to entering the work force or reentering school after a period of absence outweigh the nonpecuniary costs. On the other hand, Table 3 reveals a positive state dependence associated with marriage. The nonpecuniary divorce cost can be interpreted in thousands of 1983 dollars since the budget constraint is in thousands of 1983 dollars. This means that there is a $5822 net penalty for getting divorced. This cost can be viewed as representing the legal, economic (excluding the loss of the husband's wage earnings) and psychic costs of divorce.

6.2. *Overall within sample fit*

Overall, the model fits the data reasonably well. To illustrate this, I compare the actual proportion of women who choose each choice with the proportion predicted by the model given the parameter estimates in Tables 2 and 3. I compute predicted proportions by simulating choices for every woman in all of her sample years and then average over the number of sample members. Furthermore, I simulate choices for 100 draws of the extreme value error vector (the stochastic portion of the value functions) and average over the number of draws.

Table 4 presents the actual and predicted percentages of total person-years spent in each of the four choices as well as other measures of the model's within sample fit. The dynamic model reasonably predicts the schooling and birth control choices of the sample members. Simple chi-square goodness of fit tests do not reject the null hypothesis that the actual number of total

Table 4
Actual vs. predicted choices and select measures

	Actual	Predicted
Percent of total person-years spent:		
Working	61.29%	58.47%
In school	19.85%	19.29%
Married	47.05%	52.28%
Using birth control	74.33%	72.98%
Percent of marriages ending in divorce	29.28%	28.63%
Percent of women never married	15.94%	12.00%
Average 1st marriage duration (years)	8.52	7.91
Average highest grade completed	13.56	13.35
Average work experience (years)	10.21	9.71
Average age at first marriage	22.26	21.67
Average age at first employment	18.90	19.79
Average number births per woman	1.46	1.20
Number of marriages	929	961

[51] However, I did find that increasing the upper bound on marriage duration from 2 to 5 years does not change the sign of the marriage duration coefficient. In other words, increases in marriage duration still decrease the utility associated with being married if the upper bound is 5 rather than 2 years.

person-years is the same, at the 5 percent level, as the number predicted for both choices.[52] However, the tests reject the null hypothesis that the predicted and actual numbers of total person-years are the same for marriage and working. Specifically, the model overpredicts total married person-years, predicting that more women will ever marry (albeit for shorter durations) and at a younger age than that which is observed in the data.[53] Moreover, the model underpredicts total working person-years, underpredicting the average years of work experience and overpredicting the average age at first employment.

Figures 1–4 present the actual and predicted proportions of women who choose each choice over age. Figure 1 demonstrates that the model nicely captures the positive relationship observed

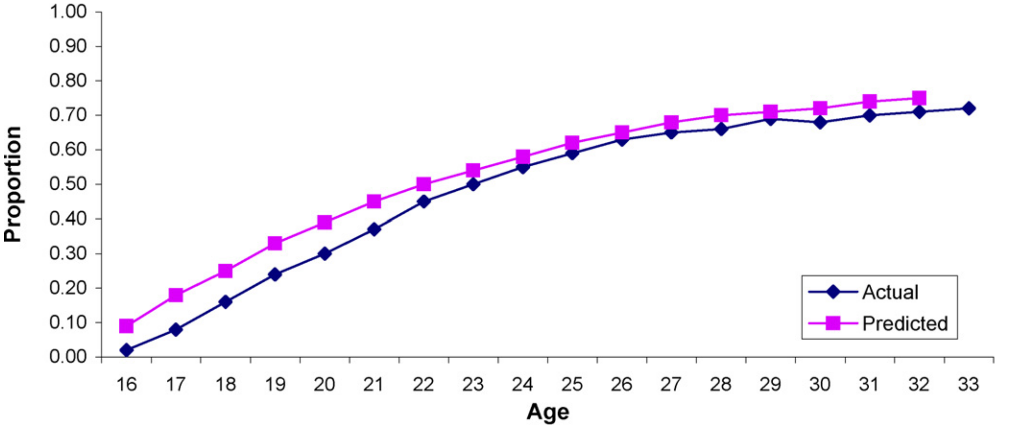

Fig. 1. Actual vs. predicted marriage proportions.

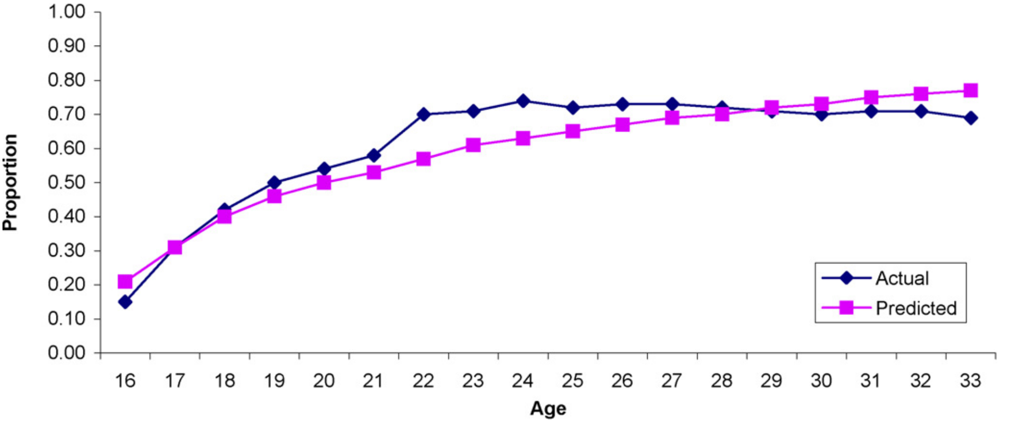

Fig. 2. Actual vs. predicted working proportions.

[52] While, $\chi^2(1, 0.95) = 3.84$, the chi-square test statistic is 20.27 for working, 81.76 for marriage, 2.54 for schooling, and 3.79 for birth control.

[53] Since the model overpredicts the number of marriages, total married person-years are overpredicted even though marriage duration is underpredicted. In addition, the percentage of marriages ending in divorce is underpredicted even though the number of divorces is slightly overpredicted.

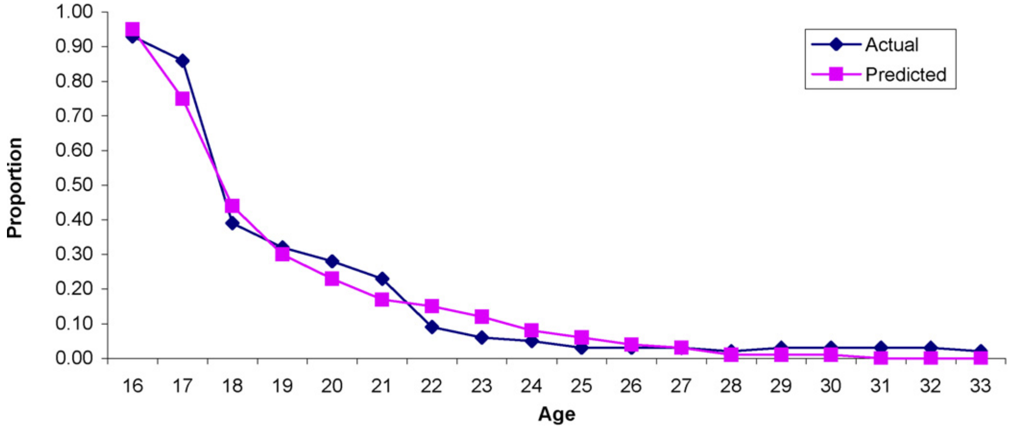

Fig. 3. Actual vs. predicted school proportions.

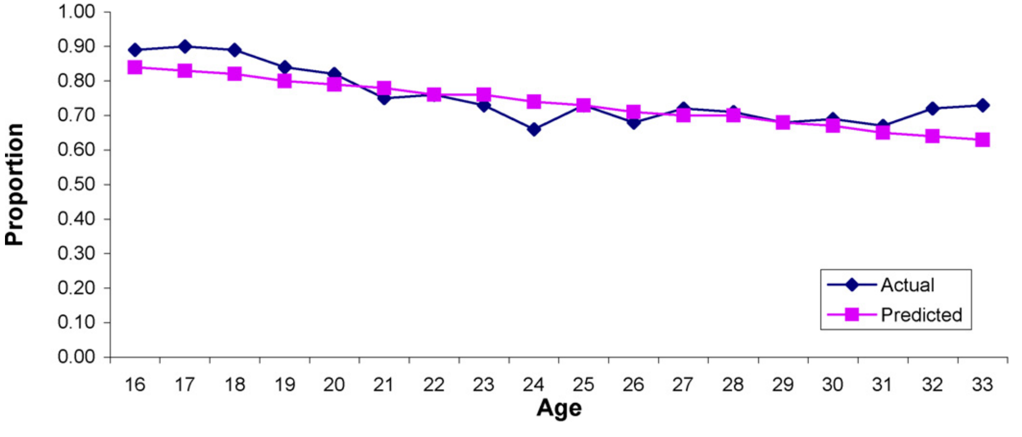

Fig. 4. Actual vs. predicted birth control proportions.

in the data between age and marriage but overpredicts marriage particularly at young ages. This may be because the model does not allow the arrival of marriage offers to depend on age or because there is no utility cost associated with the joint activity of going to school while married.[54] Figure 2 illustrates that the model also adequately captures the observed positive relationship between age and labor market work. However, it misses the substantial increase in the actual proportion of women who work around age 22 perhaps because graduation effects are ignored. Finally, Fig. 3 shows that the model performs well in predicting the proportion of women who are in school at each age, including the observed dramatic decline between the ages of 16 and 24, and Fig. 4 reveals that the model fits the overall pattern of birth control use observed in the data over age although the predicted trend is smoother than the observed trend.

Table 5 demonstrates how well the dynamic model predicts joint choices. First, the model closely predicts the percentage of person-years spent married and working (31.42 versus 31.04

[54] The fit of the model for joint activities is further discussed below.

Table 5
Select actual vs. predicted joint choices

	Actual (%)	Predicted (%)
Married and using birth control	36.47	39.67
Married and in school	1.51	4.24
Married and working	31.04	31.42
Working and in school	5.31	8.08
Working and using birth control	47.86	43.11
In school and using birth control	12.41	12.59

percent). On the other hand, the model overpredicts the percentage spent working and attending school (8.08 versus 5.31 percent) and the percentage spent married and in school (4.24 versus 1.51 percent). These results suggest the existence of important utility costs associated with joint choices. However, because there has to be a limit on the number of parameters that the model can identify with the available data, such costs are left out of the model. Lastly, the model fairly accurately predicts the percentage of person-years spent using birth control and attending school (12.59 versus 12.41 percent), overpredicts the percentage spent being married and using birth control (39.67 versus 36.47 percent), and underpredicts the percentage spent using birth control and working (43.11 versus 47.86 percent). These results are not surprising given that the model closely predicts birth control and schooling choices but overpredicts marriage and underpredicts work decisions (as was shown in Table 4 and discussed above.)

7. Policy simulations

This section uses the structural parameter estimates presented in Table 3 to predict the impact of a change in a specific parameter value on the decision paths of women. The three experiments examined here are an increase in the return to education, a decrease in the divorce cost, and a decrease in the cost of childcare. To conduct these experiments, I use the structural parameter estimates and draws of extreme value errors for each alternative to simulate choices for each sample woman in all of her sample years. Baseline choice probabilities are obtained by averaging over individuals and the number of error term draws. I then use the same sample of women and the same error draws to simulate choices after modifying the necessary parameter value to reflect the experiment at hand. Choice probabilities given this parameter change are found by averaging over individuals and error draws. Finally, the impact of the parameter value change is evaluated by comparing the new choice probabilities with the baseline probabilities. Note that because I do not model the general equilibrium effects in the labor, marriage, and education markets, I am unable to evaluate the effect of the policy change on men. This would greatly increase the computational cost of estimation and is left for future research.

In the first experiment, I increase both women's and men's return to education by 25 and 50 percent.[55] The wage parameter estimates in Table 2 indicate that an additional year of education increases women's wages by 8.4 percent and men's wages by 4.2 percent. Therefore, a 25 percent increase in the return to education yields a new return of 10.5 percent for women and 5.25 percent for men, and a 50 percent increase yields a 12.6 percent return for women and 6.3 percent

[55] Experiments in which the return to schooling was increased only for women resulted in large effects on marriage decisions because, given the increase in female wages, the value of being single more frequently dominated the value of being married.

for men. These increases are consistent with those estimated in the literature. Card (1998), for example, reports that the "conventionally-measured" return to education rose by 35 to 50 percent over the past 15 years. Ashenfelter and Rouse (1998) estimate that the return to an additional year of schooling grew by 61 percent from 1979 to 1993. Ashenfelter et al. (2000) report that the overall average return to education has been increasing by approximately 2 percentage points per decade. Given my estimated rate of return of 8.4 percent for women, this translates into a 24 percent increase in the rate of return over one decade and a 48 percent increase over two decades.

Table 6 presents the results from this experiment. Findings indicate that increasing the return to education by 25 and 50 percent increases total working years by 5.7 and 13.6 percent, respectively. Women enter the labor market at a slightly earlier age and accumulate more work experience, on average. Additionally, 25 and 50 percent increases in the return to education raise the percentage of total person-years spent in school by approximately 2.9 and 7.6 percent, respectively, increasing the average highest grade completed. Further examination reveals that such increases are only significant, at the 5 percent level, between the ages of 19 and 25. Note that whereas one might expect increases in the return to education to delay entry into the labor market as women spend more time in school, the model predicts that women start working in the labor market earlier than before while simultaneously going to school. That is, total years spent working and going to school simultaneously increase.[56]

Results further reveal that 25 and 50 percent increases in the return to education reduce total married person-years by approximately 5.0 and 18.9 percent, respectively. Figure 5 demonstrates that resulting decreases in the proportion of women who are married at each age become more pronounced as age increases, with the proportion even beginning to decline around age 27 for the 50 percent increase. Table 6 further indicates that there are fewer marriages, more divorces, more women never marry, and those who do marry younger and stay married for a shorter time. Lastly, Table 6 reveals that increasing the return to education has virtually no impact on the total years spent using birth control. However, because total married person-years fall, the average number of births per woman falls too.

In the second experiment, I decrease the cost of divorce by 25 and 50 percent. As discussed above, the divorce cost can be viewed as representing the legal, economic (excluding the loss of the husband's wage earnings) and psychic costs of divorce. Changes in the divorce cost may, therefore, simulate changes to any one of these types of costs. For instance, a reduction may reflect reforms to the legal environment, such as the switch from fault-based divorce laws to no-fault divorce laws that occurred throughout the 1970s and 1980s. This switch substantially relaxed divorce laws, making divorce less restrictive by reducing the legal obstacles and economic and psychological costs of divorce (Nakonezny et al., 1995). Quantifying these changes is difficult, and to the best of my knowledge no data exist on the costs of divorce across states. Thus, the policy experiment I conduct should be viewed as reducing the overall cost of divorce. Table 3 indicates that there is a $5822 penalty associated with getting divorced. Therefore, a 25 percent reduction in the divorce cost corresponds to approximately $1456, and a 50 percent reduction amounts to approximately $2911.

Results are presented in Table 6 and indicate that decreasing the divorce cost by 25 and 50 percent reduces the percentage of total married person-years by 12.9 and 22.1 percent, respectively. Figure 6 allows for a closer examination of the impact on the proportion of women married at

[56] As discussed in Section 6, this may be, in part, because my model does not include a utility cost associated with this joint activity.

Table 6
Policy experiments

	Baseline	Increase the return to education		Decrease the divorce cost		Decrease the cost of childcare	
		25% Increase	50% Increase	25% Decrease	50% Decrease	$2000/year	$3000/year
Percent of total person-years spent:							
Working	58.47%	61.80%	66.43%	58.94%	59.11%	64.89%	67.47%
In school	19.29%	19.84%	20.75%	19.27%	19.12%	19.15%	19.02%
Married	52.28%	49.69%	42.39%	45.52%	40.73%	58.32%	62.92%
Using birth control	72.98%	73.09%	73.22%	73.12%	73.29%	71.27%	69.90%
Percent of marriages ending in divorce	28.63%	34.50%	49.59%	57.90%	77.28%	27.49%	26.62%
Percent of women never married	12.00%	15.20%	23.61%	8.09%	3.11%	8.76%	6.59%
Average 1st marriage duration (years)	7.91	7.84	7.47	5.41	3.18	8.54	8.98
Average highest grade completed	13.35	13.44	13.59	13.35	13.32	13.33	13.31
Average work experience (years)	9.71	10.26	11.03	9.79	9.82	10.78	11.20
Average age at first marriage	21.67	21.48	21.18	21.23	20.17	21.01	20.49
Average age at first employment	19.79	19.63	19.40	19.79	19.83	19.08	18.78
Average number births per woman	1.20	1.18	1.12	1.15	1.12	1.30	1.39
Number of marriages	961	924	827	1126	1937	997	1027

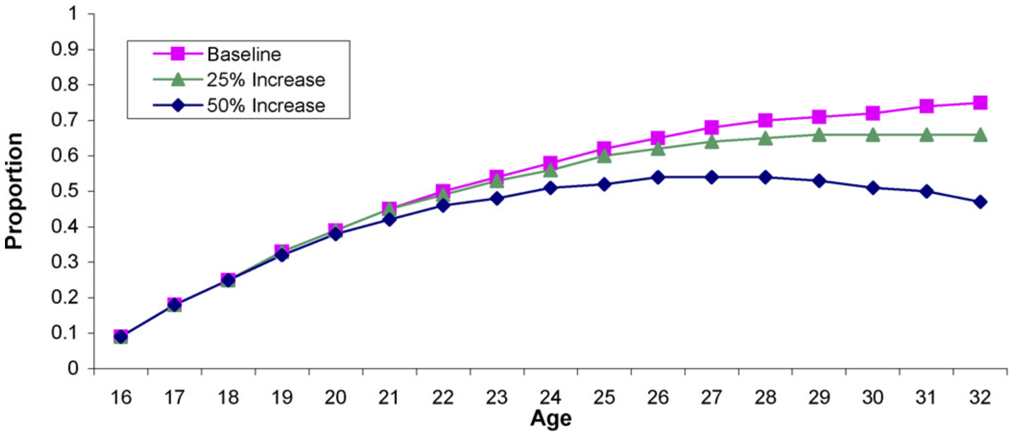

Fig. 5. Projected impact of an increase in the return to education on marriage proportions.

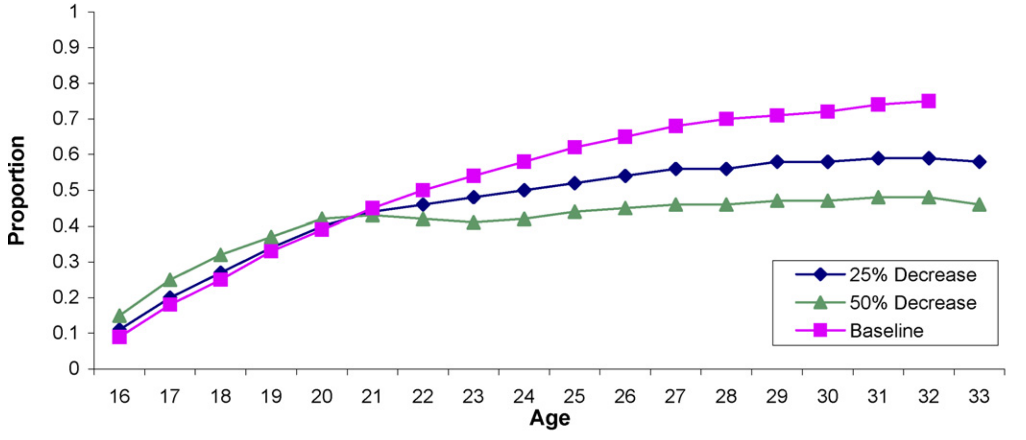

Fig. 6. Projected impact of a decrease in the divorce cost on marriage proportions.

each age and reveals that this proportion actually increases before age 21. This is because reducing the divorce cost decreases the age at first marriage and reduces the percentage of women who never marry. However, decreasing the divorce cost also increases the divorce rate substantially,[57] reducing the average marriage duration. Thus, as more and more women divorce, the proportion of women married eventually decreases below the baseline. Note that this dip occurs around the age of first marriage.[58] Moreover, lowering the divorce cost by 25 and 50 percent increases the number of marriages. The number of first marriages increases by 4.5 and 10.1 percent, respectively, while the number of subsequent marriages increases dramatically by 3 and 8 fold. This

[57] This results contradict the findings of Brien et al. (2006), who find that changing divorce laws has relatively little impact on the number of people who divorce. Friedberg (1998), on the other hand, finds that the change in divorce laws to allow for unilateral divorce accounts for 17 percent of the rise in divorce rates between 1968 and 1988. Moreover, she finds that divorce rates would have been 6 percent lower in 1988 alone had divorce laws not been changed.

[58] A blown up version of Fig. 6 reveals that this dip below the baseline happens at an earlier age for the 25 percent decrease in the divorce cost than for the 50 percent reduction.

is because when the divorce cost is lowered, a bad shock to marriage one period can induce a couple to divorce only to remarry the next period.

Surprisingly, reducing the divorce cost has relatively little impact on women's labor market work, schooling, and birth control decisions. Total working years and total years using birth control slightly increase, while total years in school slightly decrease. Lastly, the number of births declines as well, primarily because total married years fall.

In the third experiment, I simulate a reduction in the cost of childcare by decreasing the disutility mothers of children under the age of 6 feel when working in the labor market.[59] Specifically, I simulate \$2000 per year and \$3000 per year subsidies by changing the parameter in Table 3 on 'working with kids age $<$ 6 years old' from -1.206 to $+0.794$ and $+1.794$, respectively. According to the US Bureau of the Census (1995), the average family with a child of preschool age spent \$64 per week (approximately \$3300 per year) on childcare in 1986 and \$79 per week (approximately \$4100 per year) in 1993. Therefore, the changes in the cost of "childcare" (in 1983 dollars) that I simulate represent substantial reductions in the actual cost of childcare in the United States.

To further understand the magnitudes of these policies, consider the following two programs studied by Berger and Black (1992). First, Louisville's 4C (Community Coordinated Child Care) day care subsidy program pays child care centers up to \$50 per week per child depending on family income. Kentucky's Title XX Purchase of Care Subsidy Program pays centers up to \$40 per week per child, depending on family income. These amounts correspond to approximately \$ 2600 and \$2080 per child per year. Additionally, the Dependent Care Tax Credit can be used to offset up to 30 percent of child care expenses, up to \$2400 per year for a family with one child. Thus, the magnitudes of my policy simulations seem reasonable.

Results presented in Table 6 indicate that reducing the cost of childcare by \$2000 and \$3000 per year increases the overall working person-years by 11 and 15.4 percent, respectively, increasing the average number of years of work experience.[60] Figure 7 indicates that these increases in labor market work occur most among women in their late teens and early twenties. Moreover, the average age at first employment falls. This, in part, leads to a reduction in the total years in school, decreasing the average highest grade completed.

The \$2000 and \$3000 annual reductions in the cost of childcare also increase total married years by 11.6 and 20.4 percent, respectively. Moreover, fewer women never marry, and those who do marry younger, stay married longer and divorce less frequently. There are also more marriages. These findings follow since being married increases the birth probability and reducing the cost of childcare makes having children less expensive. Similarly, results show that reducing the cost of childcare decreases the overall person-years that women use birth control. This reduction

[59] This experiment does not hinge on whether working mothers actually pay for childcare. Although this information is available in the NLSY, it is not incorporated in this experiment. Instead, a childcare subsidy is given to all working mothers of children under the age of 6. This can be interpreted as either a financial payment or some nonpecuniary effort (such as increased work flexibility) intended to reduce the disutility of working. Moreover, the disutility of working is reduced only for mothers of children younger than age 6, because children older than 5 are typically in school and, consequently, have different childcare needs.

[60] Some authors, such as Ribar (1995), find married women's labor force participation to be relatively insensitive to changes in the cost of childcare. On the other hand, Connelly (1992) estimates that a 50 percent childcare subsidy increases married women's labor force participation by approximately 9 percent. Han and Waldfogel (1998) similarly find that a 50 percent childcare subsidy increases the labor force participation of married women by 10 percent and of single women by 15 percent.

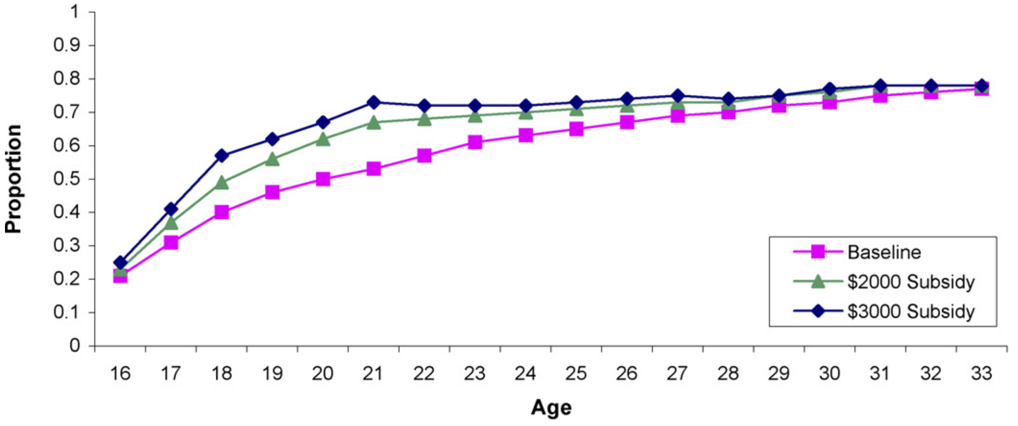

Fig. 7. Projected impact of a decrease in the cost of childcare on working proportions.

in total birth control years together with the increase in total married years increases the average number of births per woman.

8. Conclusions

In this paper, I develop and estimate a dynamic, structural model of women's labor market participation, schooling, marriage, and fertility decisions. This work provides a framework in which to analyze the career and family decisions women make over their lifetimes. Estimates of the model indicate that the career and family decisions of women vary significantly over their observable characteristics including race, religion, age, and AFQT score and strongly depend on the presence of children. For example, results suggest that young children have a bonding influence on marriage and that mothers get more utility from using birth control than women without children. This work also identifies several important sources of duration and state dependence in women's choices over time. The estimates imply a positive duration dependence in the preferences for labor market work and schooling and a negative duration dependence in the preference for marriage. Moreover, the nonpecuniary benefits of entering the work force or reentering school after a period of absence outweigh the nonpecuniary costs.

Given these estimates, I test the fit of the model to the data by comparing predicted and actual choices over all sample years and in each individual sample year. Results reveal that the dynamic model fits the data well for women's overall schooling and birth control choices. However, the model underpredicts the percentage of person-years in which women work in the labor market and overpredicts the percentage of total years women spend married.

The structural model and its parameter estimates are then used to conduct several counterfactual experiments in which the impact of a change in a specific parameter value on individuals' decision paths is measured. This paper discusses the results of three such experiments with policy relevance in which I simulate an increase in the return to education, a decrease in the divorce cost, and a reduction in the cost of childcare. As expected, results suggest that women would stay in school longer, work more in the labor market, and spend a smaller fraction of their lives married if the return to education was increased. Moreover, women would work in the labor market more, spend a greater fraction of their lives married, and use birth control less frequently if the cost of childcare was decreased. These findings illustrate the importance of modeling women's career and family choices simultaneously; a policy program targeting one choice may have important influences on the other choices. Ignoring the interdependences between women's career and family choices could, therefore, lead to incomplete forecasts of some policy change. However, results also demonstrate that some career and family choices may be invariant to certain policy changes. For example, while decreasing the divorce cost substantially impacts marriage choices, it has relatively little impact on women's labor market work, schooling, and birth control decisions. Thus, it cannot be concluded, a priori, that a change in one career or family choice will necessarily impact all other career and family choices.

Finally, I make a number of modeling assumptions to ease the high computational cost of estimating the model. While these assumptions are necessary in order to limit both the size of the state space and the number of times the value functions need to be calculated, it is important to recognize their implications.[61] For instance, I assume that all women have a marriage offer each period rather than allowing offers to depend upon observable characteristics and arrive with uncertainty. To understand the consequence of this assumption, suppose that having a child while single reduces the probability of receiving a marriage offer. Because this effect is not explicitly modeled, it is captured in the utility flow women derive from children while married, and the effect of children on the utility flow of being married is biased downward. In addition, only the age of the youngest child is tracked for estimation purposes. Therefore, the model cannot capture complicated birth spacing decisions. As a final example, the model assumes that all labor market work is full-time. As a result, policy experiments in which market work becomes more attractive underestimate the true impact on labor market participation because the model does not capture individuals who would move from no work to part-time work but for whom the increase is not enough to induce full-time work. Despite such potentially important implications, relaxing these assumptions would substantially increase the computational cost of estimating the model and is therefore left for future work.

Acknowledgment

I would like to thank Steven Stern, Michael Brien, William Johnson, Christopher Swann, Neil Seftor, two anonymous referees, and seminar participants at the University of Virginia for useful comments and suggestions on this paper. All remaining errors are my responsibility.

Appendix A

Data on birth control choices are collected for only 9 out of 17 possible sample years. Therefore, it is necessary to simulate contraception choice for years in which data are not obtained. This is accomplished using a technique similar to that of Engers and Stern (2002).

Consider a sample period in which birth control data are not collected. Given an agent's observed marriage, work, and school decisions, let choice 1 denotes her choice, $k \in K = \{p_t, m_t, b_t, e_t\}$, if she uses birth control and choice 2 her choice if she does not use birth control. Furthermore, let $\overline{V}_1$ denote the deterministic portion of the value function associated with

[61] Some of these implications have already been noted above. Footnote 18 discusses how not incorporating unobserved heterogeneity into the model overestimates the duration effects; footnote 39 discusses what would happen to the parameter estimates if the upper bounds placed on marriage duration, work experience, and education were relaxed; and Section 6 discusses the impact on results of the model's inability to capture the paternity of children.

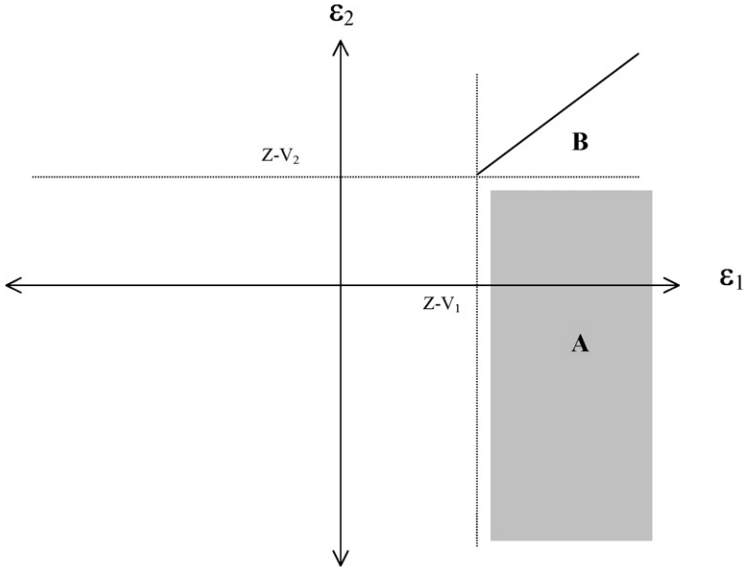

Fig. A.1.

choice 1, and $\overline{V}_2$ the deterministic part of the value function associated with choice 2. In addition, assume that Z is the value function of the best choice among the remaining 14 choices.

In Fig. A.1, the probability that birth control is used equals the sum of areas A and B. Area A is the probability of drawing an ϵ such that choice 1 is better than choice 2, and choice 2 is worse than Z. Specifically,

$$A = \int_{Z-\overline{V}_1}^{\infty} \int_{-\infty}^{Z-\overline{V}_2} f(\epsilon_1) f(\epsilon_2)\, d\epsilon_2\, d\epsilon_1. \tag{A.1}$$

Area B is the probability of drawing an ϵ such that choice 1 is better than Z, conditional on choice 2 being better than Z. Specifically,

$$B = \frac{F(\overline{V}_1 - \overline{V}_2 + \epsilon_1) - F(Z - \overline{V}_2)}{1 - F(Z - \overline{V}_2)}. \tag{A.2}$$

The probability that birth control is used is the sum of areas A and B.

In summary, the probability that birth control is used in a period is simulated by taking the following steps:

(a) Draw values for $\epsilon_3, \ldots, \epsilon_{16}$ from independent extreme value distributions;
(b) Evaluate $V_k = \overline{V}_k + \epsilon_k$ for $k = 3, 16$, and find $Z = \max_{k>2} V_k$;
(c) Evaluate area A;
(d) Draw ϵ_1 from a truncated extreme value distribution;
(e) Evaluate area B;
(f) Sum areas A and B;

(g) Repeat steps (a)–(f) R times. The probability that birth control is used is then:

$$\Pr(bc_t = 1) = \frac{1}{R}\sum_{r=1}^{R}(A + B). \tag{A.3}$$

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